The Heston Model: Theory, Simulation and Calibration

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Contents

Abstract

This seminar paper deals with the Heston model. We start from the mathematical basis and then continue to the concrete numerical implementation in the project. First we consider the CIR process, because this is exactly the variance process in the Heston model. After that, the numerical methods used in the project are introduced, namely Full-Truncation Euler, Log-Euler and Monte Carlo. In the last part we then explain how the calibration to SPY option data was implemented in the program code and which tests make the implementation plausible. Thus, both the theoretical statements and the actual code lines and figures from the project are used.

1 Introduction and Objective

The Heston model is a stochastic volatility model. The basic idea is therefore that volatility is not constant, but fluctuates randomly itself. We consider the dynamics under the risk-neutral measure

$$dS_t = rS_t\,dt + \sqrt{V_t}S_t\,dW_t^S, \tag{1}$$

$$dV_t = \kappa(\theta - V_t)\,dt + \xi\sqrt{V_t}\,dW_t^V, \tag{2}$$

where the two Brownian motions satisfy $\langle W^S, W^V\rangle_t = \rho t$. The process S_t describes the stock price and V_t the random variance.

The model is mainly interesting because it can represent smile and skew effects which do not occur in the Black–Scholes model with constant volatility [7, 3]. At the same time, however, a mathematical problem appears: the variance equation contains $\sqrt{V_t}$. This function is not globally Lipschitz at 0. Therefore one cannot simply cite only the standard theory for Lipschitz SDEs, but has to look more carefully at the CIR process.

In this paper we follow four questions:

- First: Is the variance process mathematically well defined and does it stay non-negative?
- Then: Which discretization can be used for this process in a reasonable way?
- After that: What does the concrete implementation in the project look like?
- And finally: How well do calibration and tests work on the SPY data?

2 Model Framework and Parameter Interpretation

The parameters of the Heston model are

$$(\kappa, \theta, \xi, \rho, v_0).$$

Here κ is the speed at which the variance moves back towards its long-run level. This long-run level is θ. The parameter ξ measures how strongly the variance itself fluctuates. The correlation ρ connects the price shock and the variance shock. Finally, v_0 is the initial value of the variance.

In the program code these restrictions are not only assumed theoretically, but are also checked explicitly. Thus one must have $\kappa, \theta, \xi > 0$, $\rho \in (-1, 1)$ and $v_0 \geq 0$. This is meant to prevent the optimizer from trying parameters which no longer describe a meaningful Heston model.

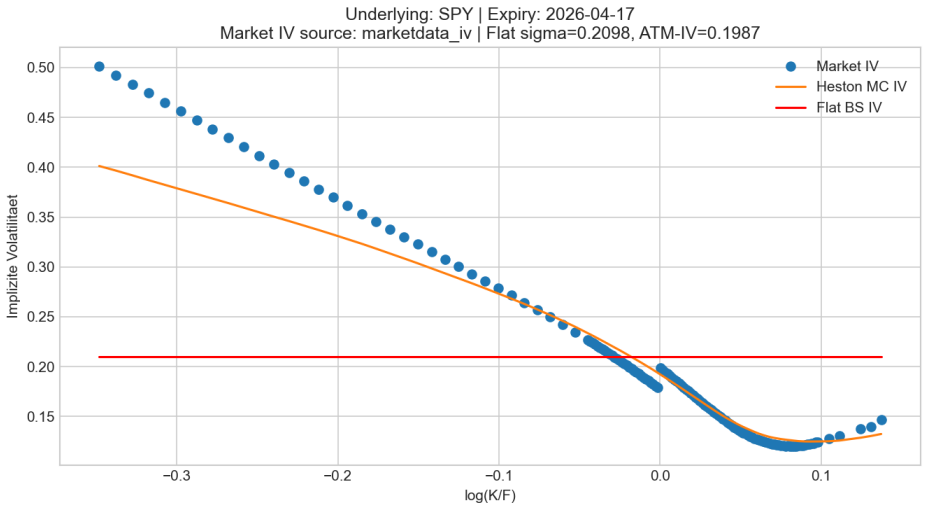


Figure 1: Comparison of the market smile with the Heston MC IV for SPY and the expiry 2026-04-17.

3 Mathematical Properties of the Variance Process (CIR)

Before we use the simulation of the Heston model, we first have to understand the variance process. This process is a CIR process and has the form

$$dV_t = \kappa(\theta - V_t)dt + \xi\sqrt{V_t}\, dW_t, \qquad V_0 \geq 0, \quad \kappa, \theta, \xi > 0.$$

The problem here is not the linear drift term, but the diffusion $\sqrt{V_t}$. At 0 this function is not globally Lipschitz. Therefore the classical existence and uniqueness theory for SDEs does not apply directly. For this reason we now consider squared Bessel processes. They

are exactly the right tool, because the CIR process can be reduced to such a process by a deterministic transformation.

3.1 Squared Bessel Processes

Definition 3.1 (Squared Bessel process)**.** *A process $(X_t)_{t\geq 0}$ is called a squared Bessel process of dimension $\delta \geq 0$, shortly* BESQ(δ)*, if it satisfies the SDE*

$$dX_t = \delta\, dt + 2\sqrt{X_t}\, dW_t, \qquad X_0 \geq 0.$$

The corresponding Bessel process is $R_t = \sqrt{X_t}$. Formally it satisfies

$$dR_t = \frac{\delta - 1}{2R_t}\, dt + dW_t.$$

Theorem 3.2 (Existence, uniqueness and non-negativity of BESQ(δ))**.** *For every $\delta \geq 0$ and $X_0 \geq 0$, the equation*

$$dX_t = \delta dt + 2\sqrt{X_t}\, dW_t$$

has a unique strong solution. Moreover, $X_t \geq 0$ for all $t \geq 0$ almost surely.

Proof source. A complete proof can be found in Revuz–Yor, *Continuous Martingales and Brownian Motion*, Chapter XI, Theorem 1.1, and in Karatzas–Shreve, *Brownian Motion and Stochastic Calculus*, Chapter 5.5.

Theorem 3.3 (Boundary classification of BESQ(δ))**.** *Let*

$$\tau_0 := \inf\{t \geq 0 : X_t = 0\}.$$

Then, for $X_0 > 0$, the following holds:

- *If $\delta \geq 2$, then the boundary 0 is not reached, that is $\mathbb{P}(\tau_0 < \infty) = 0$.*
- *If $0 < \delta < 2$, then the boundary 0 is reached almost surely, that is $\mathbb{P}(\tau_0 < \infty) = 1$.*
- *If $\delta = 0$, then 0 is reached and is absorbing.*

Proof source. See Revuz–Yor, Chapter XI, Theorem 1.2 and the following discussion. Another presentation can be found in Jeanblanc–Yor– Chesney, *Mathematical Methods for Financial Markets*, Chapter 6.

Lemma 3.4 (Yamada–Watanabe)**.** *If weak existence and pathwise uniqueness hold for an SDE, then a unique strong solution exists.*

Proof source. Yamada–Watanabe, *On the uniqueness of solutions of stochastic differential equations*, J. Math. Kyoto Univ. (1971); see also Karatzas–Shreve, Chapter 5.2.

3.2 Existence and Uniqueness for the CIR Process

Theorem 3.5 (Existence and uniqueness of the CIR process)**.** *Let $\kappa, \theta, \xi > 0$ and $V_0 \geq 0$. Then the CIR SDE*

$$dV_t = \kappa(\theta - V_t)dt + \xi\sqrt{V_t}\, dW_t$$

has a unique strong solution. In addition, $V_t \geq 0$ for all $t \geq 0$ almost surely.

Proof. We want to reduce the CIR process to a squared Bessel process. For this purpose we set

$$\delta := \frac{4\kappa\theta}{\xi^2}, \qquad c := \frac{\xi^2}{4\kappa}, \qquad \varphi(t) := c(e^{\kappa t} - 1).$$

The function φ is deterministic and strictly increasing, because

$$\varphi'(t) = \frac{\xi^2}{4}e^{\kappa t} > 0.$$

Now let X be a BESQ(δ) process with initial value $X_0 = V_0$. We define

$$V_t := e^{-\kappa t}X_{\varphi(t)}.$$

We now show that this V_t satisfies the CIR equation. By Ito's formula and the deterministic time-change rule,

$$dX_{\varphi(t)} = \delta\, \varphi'(t)dt + 2\sqrt{X_{\varphi(t)}}\, dW_{\varphi(t)}.$$

Therefore

$$dV_t = -\kappa V_t\, dt + e^{-\kappa t}\Big(\delta\, \varphi'(t)dt + 2\sqrt{X_{\varphi(t)}}\, dW_{\varphi(t)}\Big).$$

Now we only have to normalize the stochastic term correctly. We set

$$M_t := W_{\varphi(t)}$$

and consider the filtration $\mathcal{G}_t := \mathcal{F}_{\varphi(t)}$. For $0 \leq s \leq t$,

$$\mathbb{E}[M_t \mid \mathcal{G}_s] = \mathbb{E}[W_{\varphi(t)} \mid \mathcal{F}_{\varphi(s)}] = W_{\varphi(s)} = M_s.$$

Thus M is a continuous martingale. We now define

$$B_t := \int_0^t e^{-\kappa s/2}\, dM_s.$$

Then

$$\langle B\rangle_t = \int_0^t e^{-\kappa s}\varphi'(s)ds = \frac{\xi^2}{4}t.$$

By Levy's characterization,

$$\widetilde{W}_t := \frac{2}{\xi} B_t$$

is a Brownian motion. Moreover,

$$\sqrt{X_{\varphi(t)}} = e^{\kappa t/2}\sqrt{V_t}.$$

Inserting this into the equation for dV_t gives

$$dV_t = -\kappa V_t\, dt + \frac{\delta\xi^2}{4} dt + \xi\sqrt{V_t}\, d\widetilde{W}_t.$$

Since

$$\frac{\delta\xi^2}{4} = \frac{4\kappa\theta}{\xi^2} \cdot \frac{\xi^2}{4} = \kappa\theta,$$

we obtain

$$dV_t = \kappa(\theta - V_t)dt + \xi\sqrt{V_t}\, d\widetilde{W}_t.$$

Thus we have constructed a weak solution.

It remains to prove uniqueness. Let V^1 and V^2 be two solutions driven by the same Brownian motion. We show that both become BESQ processes after the inverse transformation. For this we define

$$Y_t^i := e^{\kappa t} V_t^i, \qquad X_u^i := Y_{t(u)}^i, \qquad t(u) := \frac{1}{\kappa}\log\Big(1 + \frac{4\kappa u}{\xi^2}\Big).$$

By the product rule,

$$dY_t^i = \kappa\theta e^{\kappa t} dt + \xi e^{\kappa t/2}\sqrt{Y_t^i}\, dW_t.$$

The time-change rule gives

$$dW_{t(u)} = \sqrt{t'(u)}\, d\widehat{W}_u, \qquad t'(u) = \frac{4}{\xi^2} e^{-\kappa t(u)}.$$

Therefore

$$\begin{aligned} dX_u^i &= \kappa\theta e^{\kappa t(u)} t'(u)\, du + \xi e^{\kappa t(u)/2}\sqrt{X_u^i}\, dW_{t(u)} \\ &= \frac{4\kappa\theta}{\xi^2}\, du + 2\sqrt{X_u^i}\, d\widehat{W}_u \\ &= \delta\, du + 2\sqrt{X_u^i}\, d\widehat{W}_u. \end{aligned}$$

Thus X^1 and X^2 solve the same BESQ(δ) SDE with the same initial value and the same Brownian motion. By pathwise uniqueness for BESQ we get

$$X^1 \equiv X^2.$$

Transforming back gives

$$V^1 \equiv V^2.$$

Together with Yamada–Watanabe this yields strong existence and strong uniqueness.

Finally, non-negativity follows as well. Indeed,

$$V_t = e^{-\kappa t} X_{\varphi(t)},$$

where $e^{-\kappa t} > 0$ and $X_{\varphi(t)} \geq 0$. Hence $V_t \geq 0$ for all $t \geq 0$ almost surely. □

3.3 Feller Condition for the CIR Process

Theorem 3.6 (Feller condition for CIR)**.** *Let V be the CIR process and let $V_0 > 0$. Then:*

- *If $2\kappa\theta \geq \xi^2$, then $\mathbb{P}(\inf_{t>0} V_t > 0) = 1$.*
- *If $2\kappa\theta < \xi^2$, then V reaches the boundary 0 almost surely, but remains non-negative.*

Proof. From the previous proof we know that

$$V_t = e^{-\kappa t} X_{\varphi(t)}$$

with $X \sim \mathrm{BESQ}(\delta)$ and

$$\delta = \frac{4\kappa\theta}{\xi^2}.$$

Since $e^{-\kappa t} > 0$, for every $t \geq 0$ one has

$$V_t = 0 \iff X_{\varphi(t)} = 0.$$

If we set

$$\tau_0^V := \inf\{t \geq 0 : V_t = 0\}, \qquad \tau_0^X := \inf\{u \geq 0 : X_u = 0\},$$

then the strict monotonicity of φ gives

$$\tau_0^V = \varphi^{-1}(\tau_0^X).$$

So V reaches the boundary 0 exactly when the corresponding BESQ process reaches the boundary 0.

Now we use the boundary classification of the BESQ process. If

$$\delta \geq 2,$$

then 0 is not reached. This condition is equivalent to

$$\frac{4\kappa\theta}{\xi^2} \geq 2 \iff 2\kappa\theta \geq \xi^2.$$

If, on the other hand, $0 < \delta < 2$, that is

$$0 < 2\kappa\theta < \xi^2,$$

then the boundary 0 is reached almost surely. This proves exactly the Feller condition. □

Remark 3.7 (Intuition for the Feller condition)**.** *The condition*

$$2\kappa\theta \geq \xi^2$$

means intuitively that the pull back towards the long-run level is strong enough to keep the random fluctuations of the variance away from the boundary 0*. If this condition is violated, then the fluctuations are relatively strong. In this case the process can hit the boundary* 0*. Non-negativity, however, is not lost.*

4 Numerical Methods and Convergence

4.1 Discretization: Full-Truncation Euler and Log-Euler

After the variance process has been considered mathematically, it has to be discretized for the simulation. In the project the Full-Truncation Euler scheme is used for V_t. For the price S_t, however, not S_t itself is updated directly, but $\log(S_t)$. The central code excerpt is shown in Listing 1.

Listing 1: FTE and Log-Euler step directly from the project program.

```
def fte_variance_step(
    v: np.ndarray,
    kappa: float,
    theta: float,
    xi: float,
    dt: float,
    z_v: np.ndarray,
) -> np.ndarray:
    """
    Ein Schritt des Full-Truncation-Euler-Schemas fuer die
       Varianz.

    Formel (PDF Kapitel 4.2):
```

```
        v_{n+1} = v_n + kappa (theta - v_n^+) dt + xi * sqrt(v_n^+) * sqrt(dt) * z_v
    wobei v_n^+ = max(v_n, 0) komponentenweise.
    """

    v_pos = np.maximum(v, 0.0)
    return v + kappa * (theta - v_pos) * dt + xi * np.sqrt(v_pos) * math.sqrt(dt) * z_v


def log_euler_price_step(
    log_s: np.ndarray,
    v_pos: np.ndarray,
    r: float,
    dt: float,
    z_s: np.ndarray,
) -> np.ndarray:
    """
    Log-Euler-Update fuer den Aktienpreis.

    Formel (PDF Kapitel 4.2):
        log S_{n+1} = log S_n + (r - 0.5 * v_n^+) dt + sqrt(v_n^+ * dt) * z_s

    Diese Form sichert S_{n+1} > 0 und ist numerisch stabiler als ein
    direktes Euler-Update auf S.
    """

    return log_s + (r - 0.5 * v_pos) * dt + np.sqrt(v_pos * dt) * z_s
```

The idea behind Full Truncation is simple: if a negative variance value would occur in one step, then this negative value is not used for drift and diffusion. Instead one uses

$$V_n^+ = \max(V_n, 0).$$

Thus the square-root term is always well defined. For the price, the log update ensures that after transforming back one always has $S_t > 0$.

4.2 Monte Carlo Convergence

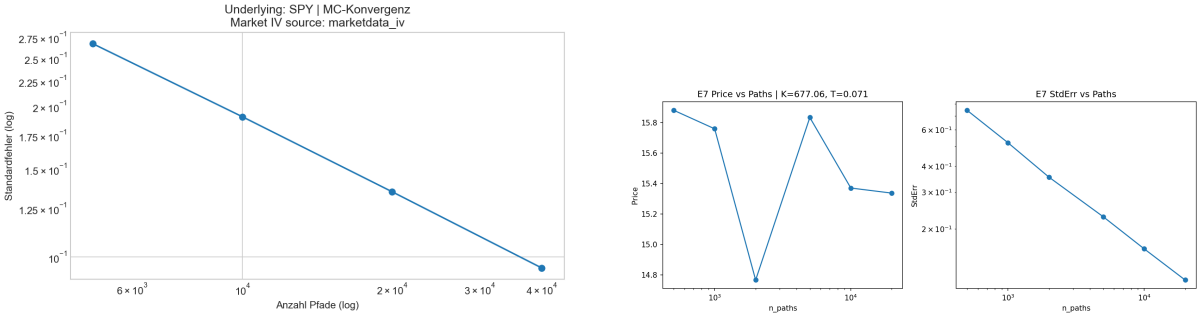
Theorem 4.1 (Monte Carlo Rate)**.** *Let $Y \in L^2$ and $\widehat{\mu}_N := \frac{1}{N}\sum_{i=1}^{N} Y_i$ with i.i.d. copies Y_i. Then*

$$\mathbb{E}\Big[|\widehat{\mu}_N - \mu|^2\Big] = \frac{\mathrm{Var}(Y)}{N}, \qquad \sqrt{N}(\widehat{\mu}_N - \mu) \Rightarrow \mathcal{N}(0, \mathrm{Var}(Y)).$$

In particular, the statistical error is of order $N^{-1/2}$.

Source. Standard result of Monte Carlo theory, see Glasserman [6].

We therefore expect that the Monte Carlo error behaves approximately like $1/\sqrt{N}$. This is exactly what validation test E7 checks. There the estimated slope is -0.5098, which is very close to the theoretical value -0.5. Thus we see that the statistical part of the simulation has the expected behaviour.



(a) MC convergence in the pricing workflow (b) Validation E7 in a log-log diagram

Figure 2: Convergence of the Monte Carlo estimate.

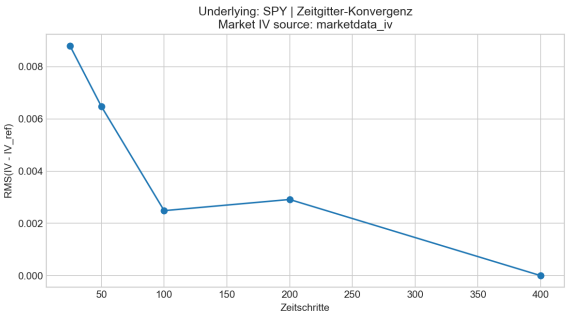
4.3 Time-Step Error and SDE Discretization

Theorem 4.2 (Euler–Maruyama under a global Lipschitz assumption)**.** *For an SDE with globally Lipschitz continuous drift and diffusion, Euler–Maruyama has strong order* $1/2$ *and weak order* 1*:*

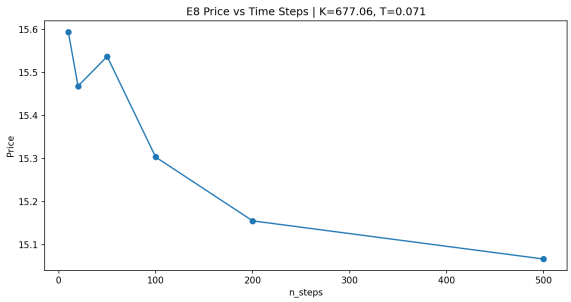
$$\sup_{0\le n\le N}\Big(\mathbb{E}|X_{t_n} - \bar{X}_n|^2\Big)^{1/2} = \mathcal{O}(\Delta t^{1/2}), \qquad |\mathbb{E}[f(X_T)] - \mathbb{E}[f(\bar{X}_N)]| = \mathcal{O}(\Delta t).$$

Source. Kloeden and Platen [8].

For the Heston or CIR process, the assumption of global Lipschitz continuity is not satisfied. Therefore Full Truncation is important. In the project, test E8 shows that the convergence with respect to the step size is still stable. The maximum relative deviation from the finest grid is 0.03498 and is therefore below the chosen tolerance 0.07.



(a) Time-step convergence in the pricing workflow



(b) Validation E8

Figure 3: Convergence when the time grid is refined.

4.4 Simulation with Diagnostics and Martingale Correction

Now we consider the actual simulation loop. Besides the update of S_t and V_t, diagnostics are also collected there. For example, the program counts negative raw values of the variance, NaN or Inf values and zero plateaus. In addition, moment matching can be used. This ensures that the empirical discounted mean of the simulated prices is closer to the theoretical martingale behaviour.

Listing 2: Core of the simulation loop from `simulate_heston_fte`.

```
    for j in range(n_steps):
        # Pre-Truncation-Check: negative Varianzwerte vor dem
           Abschneiden
        neg_pre_trunc_count += int(np.sum(v < 0.0))

        v_prev_pos = np.maximum(v, 0.0)
        near_zero_count += int(np.sum(v_prev_pos <
           v_near_zero_threshold))
        total_checks += v_prev_pos.size

        # Zufaellige Inkremente
        if use_precomputed:
            z_v_step = z_v[:, j]
            z_perp_step = z_perp[:, j]
        else:
            z_v_step = rng.standard_normal(n_paths)
            z_perp_step = rng.standard_normal(n_paths)

        z_s_step = params.rho * z_v_step + math.sqrt(1.0 - params
           .rho ** 2) * z_perp_step

```

```
        # Varianz-Update: erst rohes FTE, dann Positivitaetsprojektion.
        # Die Projektion eliminiert negative Endwerte in finalen Outputs.
        v_raw = fte_variance_step(
            v=v,
            kappa=params.kappa,
            theta=params.theta,
            xi=params.xi,
            dt=dt,
            z_v=z_v_step,
        )
        neg_after_fte_raw_count += int(np.sum(v_raw < 0.0))
        v_next = np.maximum(v_raw, 0.0)
        neg_post_projection_count += int(np.sum(v_next < 0.0))

        # Effektive Schrittvarianz im Asset-Update:
        # deterministic midpoint nur aus dem Driftterm von v.
        # Wichtig: keine direkte Nutzung des stochastischen v_{n+1}, da dies
        # bei korrelierten Schocks einen Drift-Bias verursachen kann.
        v_eff = np.maximum(
            v_prev_pos + 0.5 * params.kappa * (params.theta - v_prev_pos) * dt,
            0.0,
        )
        s_prev = np.exp(log_s)
        log_s = log_euler_price_step(
            log_s=log_s,
            v_pos=v_eff,
            r=r,
            dt=dt,
            z_s=z_s_step,
        )
        s_new = np.exp(log_s)

        if martingale_correction:
            # Moment-Matching im Mittel: reduziert Finite-Sample-Drift des
            # diskontierten Mittelwerts, ohne Pfadordnung zu
```

```
            veraendern.
54          target_mean = float(np.mean(s_prev) * math.exp(r * dt
               ))
55          observed_mean = float(np.mean(s_new))
56          if np.isfinite(target_mean) and np.isfinite(
               observed_mean) and observed_mean > 0.0:
57              corr = target_mean / observed_mean
58              s_new = s_new * corr
59              log_s = np.log(np.maximum(s_new, 1e-300))
60              corr_abs = abs(float(corr) - 1.0)
61              martingale_corr_abs_sum += corr_abs
62              martingale_corr_max_abs = max(
                   martingale_corr_max_abs, corr_abs)
63
64      # dS-Diagnostik zur Detektion von Path-Freezing / Masking
           -Artefakten.
65      ds = s_new - s_prev
66      exact_mask = ds == 0.0
67      near_mask = np.abs(ds) <= float(
           asset_increment_near_zero_eps)
68      exact_zero_increment_count += int(np.sum(exact_mask))
69      near_zero_increment_count += int(np.sum(near_mask))
70      plateau_run = np.where(exact_mask, plateau_run + 1, 0).
           astype(np.int32)
71      plateau_max = np.maximum(plateau_max, plateau_run)
72
73      # Zustand fortschreiben
74      v = v_next
75
76      # Diagnostik: Min/Max, NaN/Inf
77      v_min = float(min(v_min, float(np.min(v))))
78      v_max = float(max(v_max, float(np.max(v))))
79      nan_count += int(np.sum(np.isnan(v))) + int(np.sum(np.
           isnan(log_s)))
80      inf_count += int(np.sum(np.isinf(v))) + int(np.sum(np.
           isinf(log_s)))
81
82      if return_paths:
83          s_paths[:, j + 1] = s_new
84      if return_variance_paths:
85          v_paths[:, j + 1] = v
```

```
86
87      s_t = np.exp(log_s)
88      v_t = v
89
90      n_increments = float(max(n_paths * n_steps, 1))
91      diagnostics = {
92          "neg_pre_trunc_count": float(neg_pre_trunc_count),
93          "neg_after_fte_raw_count": float(neg_after_fte_raw_count)
                ,
94          "neg_post_projection_count": float(
                neg_post_projection_count),
95          "v_min": float(v_min),
96          "v_max": float(v_max),
97          "v_near_zero_fraction": float(near_zero_count) / float(
                total_checks) if total_checks > 0 else 0.0,
98          "exact_zero_increment_fraction": float(
                exact_zero_increment_count) / n_increments,
99          "near_zero_increment_fraction": float(
                near_zero_increment_count) / n_increments,
100         "max_exact_zero_plateau_len": float(int(np.max(
                plateau_max)) if plateau_max.size > 0 else 0),
101         "paths_with_exact_zero_plateau_ge_2": float(np.sum(
                plateau_max >= 2)),
102         "martingale_correction_mean_abs": float(
                martingale_corr_abs_sum / float(max(n_steps, 1))),
103         "martingale_correction_max_abs": float(
                martingale_corr_max_abs),
104         "nan_count": float(nan_count),
```

5 Implementation: Data, Calibration and Pricing

5.1 Command-Line Workflow

The implementation is controlled by four commands. First, `load-data` loads the option data and cleans them. After that, `calibrate` calibrates the parameters. With `make-plots` the figures are generated and with `validate` the validation is carried out. The parser definition is shown in Listing 3.

Listing 3: Excerpt from the command-line parser in `__main__.py`.

```
1       write_json(out_log, log)
2
```

```
    # Parameter separat speichern
    out_params = _outputs_dir() / "calibrated_params.json"
    write_json(out_params, {
        "kappa": best_params.kappa,
        "theta": best_params.theta,
        "xi": best_params.xi,
        "rho": best_params.rho,
        "v0": best_params.v0,
    })

    print(f"Gespeichert: {out_log}")
    print(f"Gespeichert: {out_params}")


def make_plots_cmd(args: argparse.Namespace) -> None:
    """
    Erstellt alle geforderten Plots in outputs/figures.
    """

    if args.ticker != "SPY":
        raise ValueError("Es ist ausschliesslich der Ticker SPY
           erlaubt.")

    input_csv = _outputs_dir() / "cleaned_spy_options.csv"
    if not input_csv.exists():
        raise FileNotFoundError("cleaned_spy_options.csv fehlt.
           Bitte zuerst load-data ausfuehren.")

    params_file = _outputs_dir() / "calibrated_params.json"
    if not params_file.exists():
        raise FileNotFoundError("calibrated_params.json fehlt.
           Bitte zuerst calibrate ausfuehren.")

    cleaned = pd.read_csv(input_csv)
    with params_file.open("r", encoding="utf-8") as f:
        p = json.load(f)
    params = HestonParams(
        kappa=p["kappa"],
        theta=p["theta"],
        xi=p["xi"],
        rho=p["rho"],
```

```
        v0=p["v0"],
    )

    calibration_selected_expiries: List[str] = []
    cal_log = _outputs_dir() / "calibration_log.json"
    if cal_log.exists():
        try:
            with cal_log.open("r", encoding="utf-8") as f:
                log_payload = json.load(f)
            selected = log_payload.get("expiries_selected", [])
            if isinstance(selected, list):
                calibration_selected_expiries = [str(x) for x in
                    selected if x is not None]
        except Exception:
            calibration_selected_expiries = []

    if args.expiry:
        expiry = args.expiry[0]
    else:
        expiry = None
        if calibration_selected_expiries:
            expiry = calibration_selected_expiries[0]

        # Sekundaerer Fallback: medianes days_to_expiry.
        if not expiry:
            cleaned["days_to_expiry"] = pd.to_numeric(cleaned["
                days_to_expiry"], errors="coerce")
            summary = cleaned.groupby("expiry").agg(avg_days=("
                days_to_expiry", "mean")).reset_index()
            summary = summary.sort_values(by="avg_days")
            expiry = str(summary.iloc[len(summary) // 2]["expiry"
                ])

    results = make_all_plots(
        cleaned=cleaned,
        params=params,
        expiry=expiry,
        output_dir=_figures_dir(),
        iv_source_mode=args.iv_source,
        n_paths_final=args.n_paths_final,
        steps_per_year=args.steps_per_year,
```

```
        seed=args.seed,
        strike_for_pathplot=args.strike_for_pathplot,
        calibration_expiries=calibration_selected_expiries,
    )

    for k, v in results.items():
        print(f"{k}: {v}")
```

5.2 Calibration Function and Weights

In the calibration we look for parameters such that the implied volatilities produced by the model are as close as possible to the market IVs. Thus the objective function is a weighted error in IV space. In particular, the wings of the smile can be weighted more strongly, because this is exactly where the difference between Black–Scholes and Heston becomes visible.

Listing 4: Definition of weights in the calibration.

```
def _compute_weights(df: pd.DataFrame, mode: str, spread_eps:
   float) -> np.ndarray:
    """
    Gewichte fuer die Kalibrierung im IV-Raum.

    - mode="spread": inverse relativer Spread (mit Clipping)
    - mode="spread_moneyness": spread-Gewichte plus Moneyness-
       Balancierung
    - mode="uniform": 1
    """

    if mode == "uniform":
        return np.ones(len(df), dtype=float)

    spread = pd.to_numeric(df.get("spread"), errors="coerce")
    mid = pd.to_numeric(df.get("mid"), errors="coerce")
    rel_spread = spread / np.maximum(mid.abs(), spread_eps)
    rel_spread = rel_spread.replace([np.inf, -np.inf], np.nan).
       fillna(1.0)

    base_weights = 1.0 / np.maximum(rel_spread.to_numpy(dtype=
       float), spread_eps)
    base_weights = np.clip(base_weights, 0.2, 25.0)

```

```
    if mode == "spread":
        weights = base_weights
        weights = weights / max(float(np.mean(weights)), 1e-12)
        return weights

    if mode != "spread_moneyness":
        raise ValueError(f"Unbekannter weights_mode: {mode}")

    # Moneyness-Multiplikator: Fluegel (grosses |log(K/F)|) etwas staerker gewichten.
    K = pd.to_numeric(df.get("K"), errors="coerce").to_numpy(dtype=float)
    S0 = pd.to_numeric(df.get("S0"), errors="coerce").to_numpy(dtype=float)
    r = pd.to_numeric(df.get("r"), errors="coerce").to_numpy(dtype=float)
    T = pd.to_numeric(df.get("T"), errors="coerce").to_numpy(dtype=float)
    F = np.maximum(S0 * np.exp(r * T), 1e-12)
    abs_log_m = np.abs(np.log(np.maximum(K, 1e-12) / F))
    scale = max(float(np.nanpercentile(abs_log_m, 90)), 1e-6)
    wing_boost = 1.0 + 0.8 * (abs_log_m / scale)
    wing_boost = np.clip(wing_boost, 1.0, 2.6)

    # Dichte-Balancierung je Expiry: unterbesetzte Moneyness-Bereiche aufwerten.
    density_boost = np.ones(len(df), dtype=float)
    log_m_series = pd.Series(np.log(np.maximum(K, 1e-12) / F), index=df.index)
    for expiry, grp in df.groupby("expiry", sort=False):
        idx = grp.index
        lm = log_m_series.loc[idx]
        lm = lm.replace([np.inf, -np.inf], np.nan).dropna()
        if lm.empty:
            continue

        n_bins = int(min(10, max(4, round(math.sqrt(len(lm))))))
        lo = float(lm.min())
        hi = float(lm.max())
        if not np.isfinite(lo) or not np.isfinite(hi) or abs(hi - lo) < 1e-12:
```

```
            continue

        edges = np.linspace(lo, hi, n_bins + 1)
        bin_ids = np.digitize(lm.to_numpy(dtype=float), edges
            [1:-1], right=False)
        counts = np.bincount(bin_ids, minlength=n_bins)
        positive = counts[counts > 0]
        if positive.size == 0:
            continue
        target = float(np.median(positive))
        local = target / np.maximum(counts[bin_ids].astype(float)
            , 1.0)
        local = np.clip(local, 0.6, 2.0)
        density_boost[df.index.get_indexer(lm.index)] = local

    weights = base_weights * wing_boost * density_boost
    weights = np.clip(weights, 0.2, 40.0)
    weights = weights / max(float(np.mean(weights)), 1e-12)
```

Listing 5: Objective function and data-set evaluation per expiry.

```
    def objective_for_dataset(
        params: HestonParams,
        dataset: pd.DataFrame,
        dataset_weights: pd.Series,
        n_paths: int,
        z_cache_local: Dict[str, Tuple[np.ndarray, np.ndarray]],
    ) -> float:
        total = 0.0
        for expiry in expiries:
            df_exp = dataset[dataset["expiry"] == expiry]
            if df_exp.empty:
                continue
            n_steps = n_steps_map[expiry]

            # Simuliere mit CRN
            result = compute_model_iv_for_expiry(
                params=params,
                df_expiry=df_exp,
                n_paths=n_paths,
                n_steps=n_steps,
                seed=config.seed,
```

```
                use_precomputed=True,
                z_cache=z_cache_local[expiry],
            )

            model_iv = result["model_iv"]
            market_iv = df_exp["iv_market"].to_numpy(dtype=float)
            w = dataset_weights.loc[df_exp.index].to_numpy(dtype=
                float)

            # Fehler berechnen; NaNs mit hohem Penalty
            diff = model_iv - market_iv
            diff = np.where(np.isfinite(diff), diff, 5.0)
            total += float(np.sum(w * diff ** 2))

        return total

    def objective(x: np.ndarray) -> float:
        params = HestonParams.from_array(x)
        try:
            params.validate()
        except Exception:
            return 1e6
        return objective_for_dataset(
            params=params,
            dataset=df_opt,
            dataset_weights=weights_opt,
            n_paths=config.n_paths,
            z_cache_local=z_cache_opt,
```

Listing 6: Optimization with restarts and early stopping.

```
            res = minimize(
                objective,
                x_start,
                method="L-BFGS-B",
                bounds=bounds,
                options=min_opts if min_opts else None,
            )

            full_obj = float(res.fun)
            if subsampling_active:
                try:
```

```
                params_eval = HestonParams.from_array(res.x)
                params_eval.validate()
                full_obj = objective_for_dataset(
                    params=params_eval,
                    dataset=df_full,
                    dataset_weights=weights_full,
                    n_paths=n_paths_select,
                    z_cache_local=z_cache_select,
                )
            except Exception:
                full_obj = 1e12

        selection_obj = float(full_obj if subsampling_active else res.fun)
        selection_weighted_rmse = math.sqrt(max(selection_obj, 0.0) / selection_weight_sum)
        opt_weighted_rmse = math.sqrt(max(float(res.fun), 0.0) / opt_weight_sum)
        full_weighted_rmse = math.sqrt(max(float(full_obj), 0.0) / full_weight_sum)

        prev_best_obj = best_selection_obj
        relative_improvement = 0.0
        improved = selection_obj < best_selection_obj
        meaningful_improvement = False
        if improved:
            if np.isfinite(prev_best_obj):
                relative_improvement = (prev_best_obj - selection_obj) / max(abs(prev_best_obj), 1e-12)
            else:
                relative_improvement = float("inf")
            meaningful_improvement = (not np.isfinite(relative_improvement)) or (relative_improvement >= config.min_relative_improvement)

        if improved:
            best_selection_obj = float(selection_obj)
            best_obj_opt = float(res.fun)
            best_obj_full = float(full_obj)
            best_x = res.x
```

```
45                stagnation_count = 0 if meaningful_improvement
                      else (stagnation_count + 1)
46            else:
47                stagnation_count += 1
48
49            all_runs.append({
50                "restart": i,
51                "success": bool(res.success),
52                "fun": float(res.fun),
53                "fun_full": float(full_obj),
54                "selection_fun": float(selection_obj),
55                "weighted_rmse_subsample": float(
                      opt_weighted_rmse),
56                "weighted_rmse_full": float(full_weighted_rmse),
57                "weighted_rmse_selection": float(
                      selection_weighted_rmse),
58                "relative_improvement_vs_prev_best": float(
                      relative_improvement if np.isfinite(
                      relative_improvement) else -1.0),
59                "meaningful_improvement": bool(
                      meaningful_improvement),
60                "x": res.x.tolist(),
61                "message": str(res.message),
62                "nit": int(getattr(res, "nit", 0)),
63                "nfev": int(getattr(res, "nfev", 0)),
64            })
65
66            if bool(res.success):
67                successful_runs.append({
68                    "fun": float(res.fun),
69                    "fun_full": float(full_obj),
70                    "selection_fun": float(selection_obj),
71                    "selection_weighted_rmse": float(
                          selection_weighted_rmse),
72                    "x": res.x,
73                })
74
75            best_weighted_rmse = math.sqrt(max(best_selection_obj
                  , 0.0) / selection_weight_sum)
76            if n_restarts_executed >= int(config.
                  min_restarts_before_stop):
```

```
77                  if config.target_weighted_rmse is not None and
                        best_weighted_rmse <= float(config.
                        target_weighted_rmse):
78                      early_stop_reason = (
79                          f"target_weighted_rmse_reached: best={
                                best_weighted_rmse:.6f}, "
80                          f"target={float(config.
                                target_weighted_rmse):.6f}"
81                      )
82                      break
83                  if config.early_stop_patience > 0 and
                        stagnation_count >= int(config.
                        early_stop_patience):
84                      early_stop_reason = (
85                          "stagnation: "
86                          f"{stagnation_count} restarts ohne
                                signifikante Verbesserung "
87                          f"(min_relative_improvement={config.
                                min_relative_improvement:.4f})"
88                      )
89                      break
```

5.3 IV Inversion

To compare market prices and model prices, both are transformed into implied volatilities. For this purpose the Black–Scholes formula is inverted numerically. Before that, the program checks whether the price lies within the no-arbitrage bounds. This filters out implausible or numerically problematic values.

Listing 7: Inversion of the Black–Scholes formula with `brentq`.

```
1  def implied_vol_from_price(
2      price: float,
3      S0: float,
4      K: float,
5      T: float,
6      r: float,
7      option_type: str,
8      vol_lower: float = 1e-6,
9      vol_upper: float = 5.0,
10 ) -> ImpliedVolResult:
11     """
```

```
    Invertiert einen Black-Scholes-Preis auf implizite Volatilitaet.

    Die Funktion prueft zuerst No-Arbitrage-Grenzen. Liegt der Preis ausserhalb
    der Bounds, wird status="arbitrage_violation" gesetzt.
    """

    # Eingabepfad pruefen, bevor numerisch gesucht wird.
    if price <= 0.0 or S0 <= 0.0 or K <= 0.0 or T <= 0.0:
        return ImpliedVolResult(iv=float("nan"), status="bad_input")

    # No-Arbitrage-Grenzen fuer den Preis
    if option_type == "call":
        lower = max(S0 - K * math.exp(-r * T), 0.0)
        upper = S0
    elif option_type == "put":
        lower = max(K * math.exp(-r * T) - S0, 0.0)
        upper = K * math.exp(-r * T)
    else:
        raise ValueError("option_type muss 'call' oder 'put' sein.")

    if price < lower - 1e-12 or price > upper + 1e-12:
        return ImpliedVolResult(iv=float("nan"), status="arbitrage_violation")

    def objective(vol: float) -> float:
        return bs_price(S0, K, T, r, vol, option_type) - price

    try:
        f_low = objective(vol_lower)
        f_high = objective(vol_upper)

        # Falls kein Vorzeichenwechsel, existiert kein loesbarer Root im Intervall.
        if f_low * f_high > 0.0:
            return ImpliedVolResult(iv=float("nan"), status="no_solution")

```

```
        iv = brentq(objective, vol_lower, vol_upper, maxiter=200)
        return ImpliedVolResult(iv=float(iv), status="ok")
    except Exception:
        return ImpliedVolResult(iv=float("nan"), status="failed")
```

5.4 Calibration Result (SPY)

The parameters stored in `outputs/calibrated_params.json` are:

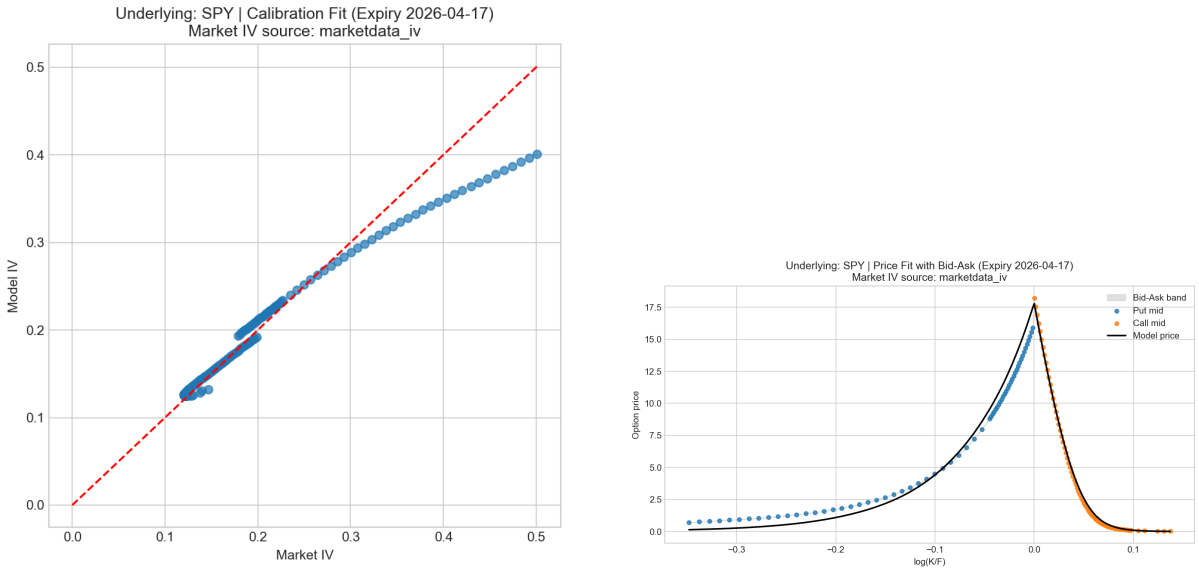
Parameter	Value
κ	1.463 129
θ	0.066 619
ξ	1.136 194
ρ	−0.853 467
v_0	0.044 775

Table 1: Calibrated Heston parameters for SPY.

Now the Feller condition can be checked directly. For this parameter set one has

$$2\kappa\theta = 0.19495 < \xi^2 = 1.29094.$$

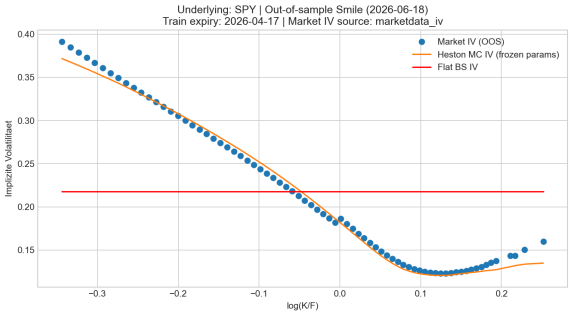
Thus the condition is not satisfied. This does not mean that the process is useless. It only means that the boundary 0 can be reached by the variance process. This fits the theoretical discussion from Chapter 3.



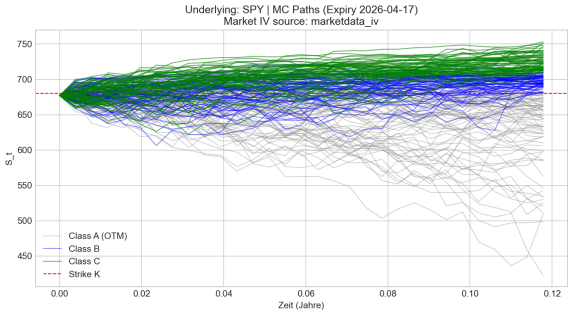
(a) Scatter plot of the IV fit

(b) Price fit with bid-ask band

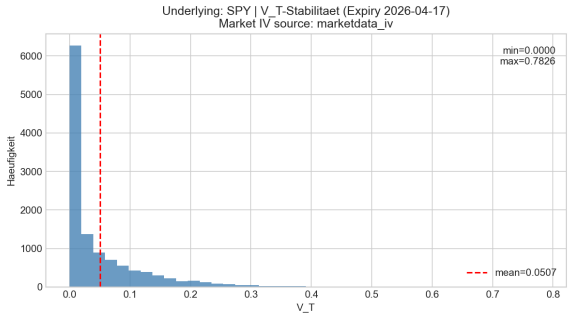
Figure 4: Calibration in IV space and price space.



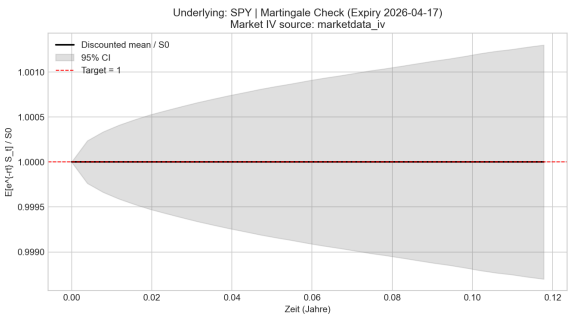
(a) Smile outside the direct calibration set



(b) MC paths by payoff classes

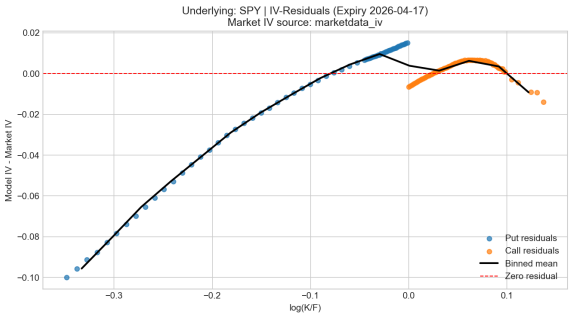


(c) Terminal variance

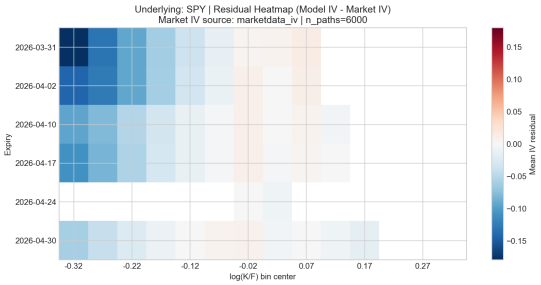


(d) Discounted martingale check

Figure 5: Further figures from the plotting workflow.



(a) IV residuals against log-moneyness



(b) Residual heatmap across expiries

Figure 6: Residuals of the calibration.

6 Validation and Plausibility Tests

After the parameters have been calibrated, we have to check whether the simulation is also numerically meaningful. The test framework in `validation.py` carries out several test blocks for this purpose. Among other things, path consistency, martingale behaviour, Brownian-increment quality, convergence, no-arbitrage checks, special cases and reproducibility are tested.

Listing 8: Test for positive and finite simulation results.

```python
"""
Tests fuer Full-Truncation-Euler und Log-Preis-Update.
"""

import numpy as np

from heston_mc.models import HestonParams
from heston_mc.pricers import simulate_heston_fte


def test_fte_simulation_positive_and_finite():
    params = HestonParams(kappa=1.5, theta=0.04, xi=0.3, rho
       =-0.5, v0=0.04)
    rng = np.random.default_rng(42)

    res = simulate_heston_fte(
        params=params,
        S0=100.0,
        r=0.01,
        T=1.0,
        n_steps=50,
        n_paths=2000,
        rng=rng,
        return_paths=False,
        return_variance_paths=False,
    )

    assert np.all(np.isfinite(res.s_t))
    assert np.all(res.s_t > 0.0)
    assert np.all(np.isfinite(res.v_t))
    assert np.all(res.v_t >= 0.0)
    assert float(res.diagnostics.get("neg_post_projection_count",
        0.0)) == 0.0
```

Listing 9: Test of the selection metric in the calibration.

```python
def test_calibration_uses_full_objective_when_subsampled():
    cleaned = _make_synthetic_cleaned()
    config = CalibrationConfig(
        n_paths=160,
        n_paths_final=500,
        steps_per_year=60,
```

```
        seed=123,
        n_restarts=1,
        weights_mode="spread",
        maxiter=1,
        max_options_per_expiry=8,
        otm_only=True,
        max_abs_log_moneyness=0.35,
    )

    _, log = calibrate_heston(cleaned=cleaned, expiries=["2026-04-17"], config=config)

    assert log["selection_metric"] == "full_objective"
    assert log["best_objective"] == pytest.approx(log["best_objective_full"])
    assert "best_objective_subsample" in log
    assert "fun_full" in log["runs"][0]


def test_calibration_uses_standard_objective_without_subsampling():
    cleaned = _make_synthetic_cleaned()
    config = CalibrationConfig(
        n_paths=120,
        n_paths_final=300,
        steps_per_year=40,
        seed=7,
        n_restarts=1,
        weights_mode="spread",
        maxiter=1,
        max_options_per_expiry=None,
        otm_only=True,
        max_abs_log_moneyness=0.35,
    )

    _, log = calibrate_heston(cleaned=cleaned, expiries=["2026-04-17"], config=config)

    assert log["selection_metric"] == "objective"
    assert log["best_objective"] == pytest.approx(log["best_objective_subsample"])
```

Listing 10: Fast end-to-end validation test.

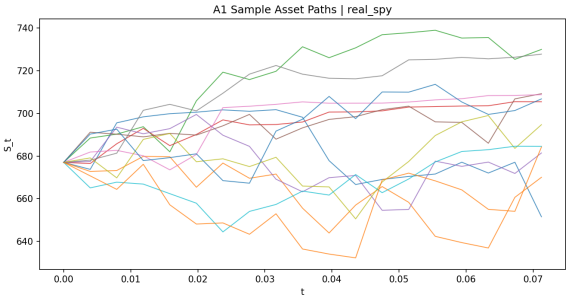
```python
def test_validation_suite_runs_synthetic_quick():
    tmp_dir = _workspace_test_tmp_dir()

    cfg = ValidationConfig(
        output_dir=tmp_dir / "diagnostics",
        seed=12345,
        steps_per_year=80,
        n_paths_main=600,
        n_paths_pricing=800,
        n_paths_market_residual=600,
        n_paths_heatmap=400,
        n_paths_convergence_steps=1200,
        n_paths_reference_fallback=1500,
        n_paths_special_case=1200,
    )
    cfg.path_count_grid = (200, 400, 800)
    cfg.step_count_grid = (10, 20, 50)
    cfg.maturities_for_pricing = (0.1, 0.25)
    cfg.strike_multipliers = (0.9, 1.0, 1.1)
    cfg.max_expiries_for_heatmap = 2

    try:
        case = make_synthetic_validation_case(name="
           synthetic_test")
        result = run_validation_suite(case=case, config=cfg)

        summary_csv = Path(result["summary_csv"])
        summary_md = Path(result["summary_md"])
        assert summary_csv.exists()
        assert summary_md.exists()
        assert result["counts"]["total"] > 0
    finally:
```

6.1 Evaluated Test Results (`real_spy`)

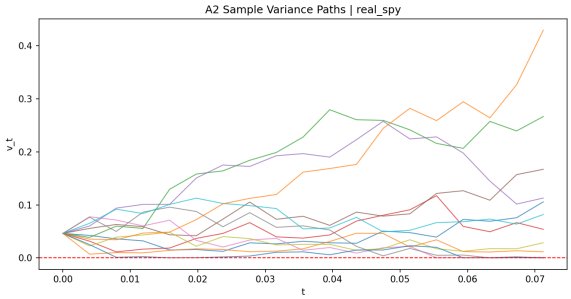
From `validation_summary.csv` we obtain 21 metrics. Of these, 20 metrics pass. Only block G11 fails narrowly, because there `iv_rmse` $= 0.03221$ is measured for a tolerance of 0.03. Thus this is not an error in the simulation itself, but shows that a single Heston parameter set can only approximate the whole market smile.

Test	Metric	Value	Result
E7_convergence_paths	slope of stderr vs paths	-0.5098	passed
E8_convergence_steps	max rel. diff. to finest grid	0.03498	passed
G10_iv_roundtrip	max absolute IV error	2.70×10^{-9}	passed
G11_residual_diagnostics	IV-RMSE	0.03221	failed
H12_static_no_arbitrage	fraction of violations	0	passed
J14_reproducibility	max absolute difference	0	passed

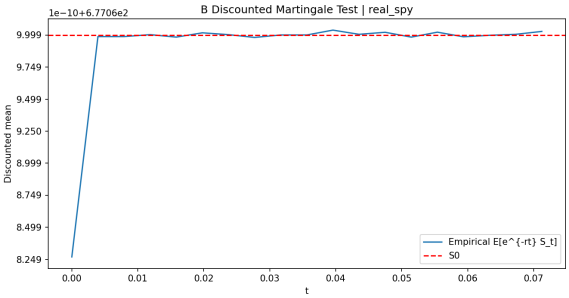
Table 2: Excerpt of central validation metrics from `real_spy`.



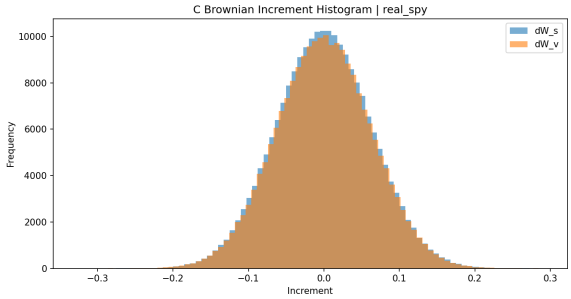
(a) A1: stock-price paths



(b) A2: variance paths

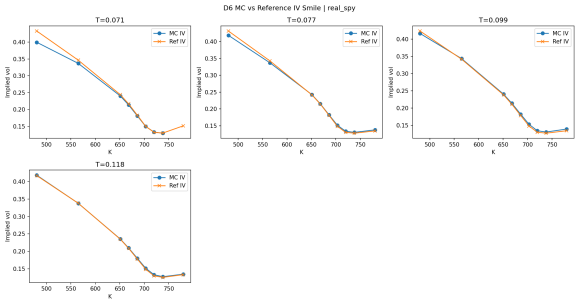


(c) B: discounted martingale

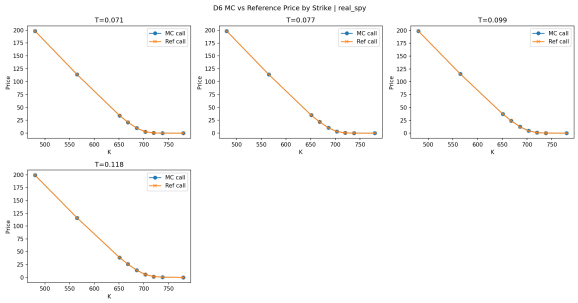


(d) C: Brownian increments

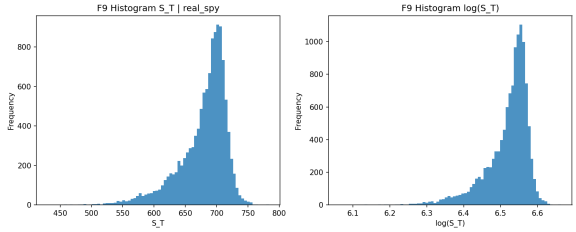
Figure 7: Plausibility figures from the test workflow.



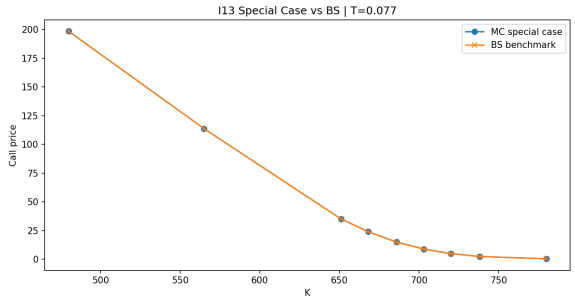
(a) D6: MC versus reference in the smile



(b) D6: MC versus reference in price

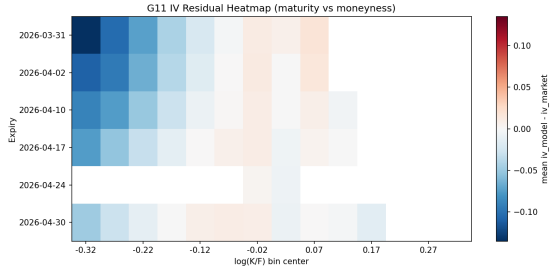


(c) F9: terminal distribution

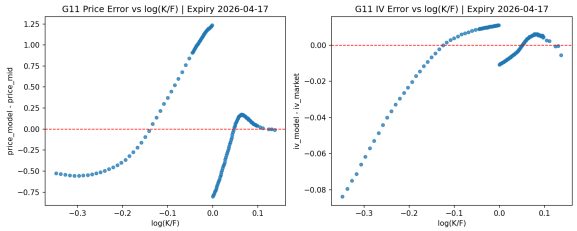


(d) I13: special case, limit to Black–Scholes

Figure 8: Further diagnostic validation figures.



(a) G11: heatmap of residuals



(b) G11: residual scatter plot

Figure 9: Residual diagnostics of the only failed block.

7 Limits of the Heston Model and Motivation for Rough Volatility

7.1 Limits of the Classical Heston Model

Now the question is why systematic residuals remain despite the calibration. The reason is not only Monte Carlo accuracy, but also the model itself. The classical Heston model has only one volatility factor and is Markovian. Empirical volatility, however, often shows strong short-term behaviour and a kind of memory which can only be captured to a limited extent by one classical factor.

Concretely, one sees three limits:

- The model has only one stochastic volatility factor.
- Short-term skews in market data are often steeper than in the classical Heston model.
- One single global parameter set can only approximately match several maturities at the same time.

7.2 Transition to Rough Volatility

One possible extension is given by rough volatility models. There the Markovian variance dynamics are replaced by a Volterra structure:

$$V_t = V_0 + \int_0^t K(t-s)\,\kappa(\theta - V_s)\,ds + \int_0^t K(t-s)\,\xi\sqrt{V_s}\,dW_s.$$

For

$$K(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}, \qquad H \in (0, 1/2),$$

a rougher volatility process is obtained [5, 2, 4]. In this way one can explain exactly the error structure which remains visible in the classical Heston model, especially for short maturities.

8 Conclusion

We have therefore considered the Heston model from its mathematical foundation to its concrete implementation. First, the CIR process was treated using BESQ results and time-change arguments. After that, we explained why Full-Truncation Euler and Log-Euler are used for the simulation. In the implementation this leads to stable paths, plausible Monte Carlo rates and a reproducible calibration to SPY options.

The validation shows that almost all numerical tests are passed. The only remaining critical point is the IV residual test. This error is, however, interesting in terms of content: it shows that the classical Heston model can capture smile structures, but does not match the market perfectly for all maturities. This also gives the motivation for rough volatility models.

A Auxiliary Results for Chapter 3

Theorem A.1 (Deterministic time-change rule)**.** *Let W be a Brownian motion and let $\tau : [0,\infty) \to [0,\infty)$ be deterministic, C^1, strictly increasing and satisfy $\tau(0) = 0$. Then*

there exists a Brownian motion $\widehat{W}$ such that, for suitable integrands H,

$$\int_0^t H_s \, dW_{\tau(s)} = \int_0^t H_s \sqrt{\tau'(s)} \, d\widehat{W}_s$$

holds. In differential notation we therefore write

$$dW_{\tau(s)} = \sqrt{\tau'(s)} \, d\widehat{W}_s.$$

Proof source. This result follows from the time change of continuous martingales, see Revuz–Yor, *Continuous Martingales and Brownian Motion*, Chapter V, or Karatzas–Shreve, *Brownian Motion and Stochastic Calculus*, Chapter 3.

Lemma A.2 (Product rule for a deterministic factor)**.** *Let X be a continuous semimartingale and let $f \in C^1([0,\infty))$ be deterministic. Then*

$$d(f(t)X_t) = f(t) \, dX_t + f'(t) X_t \, dt.$$

Proof source. This is the product rule as a special case of Ito's formula, see Karatzas–Shreve, Chapter 3.

Lemma A.3 (Normalization of a continuous martingale)**.** *Let $(M_t)_{t\geq 0}$ be a continuous local martingale with*

$$\langle M \rangle_t = ct$$

for some $c > 0$. Then

$$\widetilde{W}_t := \frac{M_t}{\sqrt{c}}$$

is a Brownian motion.

Proof source. This follows from Levy's characterization or from the Dambis–Dubins–Schwarz relation, see Revuz–Yor, Chapter V.

Definition A.4 (Pathwise uniqueness)**.** *An SDE has pathwise uniqueness if two solutions with the same initial value and the same Brownian motion are almost surely identical. Thus, for two such solutions X^1 and X^2,*

$$\mathbb{P}\Big(X_t^1 = X_t^2 \text{ for all } t \geq 0\Big) = 1.$$

Source. Yamada–Watanabe (1971), and Karatzas–Shreve, Chapter 5.2.

B Further Implementation Orchestration

Finally we consider the plot orchestrator. This part of the code does not calibrate again, but takes the stored data and parameters and creates the figures from them.

Listing 11: Plotting workflow (`make_all_plots`) directly from the project.

```python
def make_all_plots(
    cleaned: pd.DataFrame,
    params,
    expiry: str,
    output_dir: Path,
    iv_source_mode: str,
    n_paths_final: int,
    steps_per_year: int,
    seed: int,
    strike_for_pathplot: float | None = None,
    calibration_expiries: Optional[List[str]] = None,
) -> Dict[str, str]:
    """
    Orchestriert die Erstellung aller geforderten Plots.
    """

    output_dir.mkdir(parents=True, exist_ok=True)

    df_expiry = cleaned[cleaned["expiry"] == expiry].copy()
    if df_expiry.empty:
        raise ValueError("Expiry fuer Plot nicht in den Daten gefunden.")

    # Gleiche Datenbasis wie in der Kalibrierung (OTM-only + Moneyness-Cap),
    # damit Fit-Plot und Optimierungsziel konsistent sind.
    df_expiry = _canonical_plot_universe(df_expiry)
    if df_expiry.empty:
        raise ValueError("Nach OTM/Moneyness-Filter sind keine Plot-Daten uebrig.")

    S0 = float(df_expiry["S0"].iloc[0])
    r = float(df_expiry["r"].iloc[0])
    T = float(df_expiry["T"].iloc[0])

    n_steps = _n_steps_from_T(T, steps_per_year)

    # Modell-IVs fuer Plots (groessere Pfadzahl)
    result = compute_model_iv_for_expiry(
        params=params,
```

```
        df_expiry=df_expiry,
        n_paths=n_paths_final,
        n_steps=n_steps,
        seed=seed,
        use_precomputed=False,
    )

    model_iv = result["model_iv"]

    # Flat-Vol-Fit und ATM-IV
    sigma_flat = fit_flat_vol(df_expiry["iv_market"].to_numpy(
        dtype=float))
    F = forward_price(S0, r, T)
    atm_strike = select_strike_near_forward(df_expiry, F)
    atm_iv = float(df_expiry.loc[df_expiry["K"] == atm_strike, "
        iv_market"].iloc[0])

    iv_source_label = label_iv_source(iv_source_mode, df_expiry["
        iv_market_source"].unique())

    # 1) Smile Overlay
    smile_path = output_dir / f"smile_overlay_SPY_{expiry}.png"
    plot_smile_overlay(
        df_expiry=df_expiry,
        model_iv=model_iv,
        sigma_flat=sigma_flat,
        atm_iv=atm_iv,
        output_path=smile_path,
        iv_source_label=iv_source_label,
    )

    # 2) MC Path Plot
    strike = strike_for_pathplot
    if strike is None:
        strike = atm_strike

    sim_paths = simulate_heston_fte(
        params=params,
        S0=S0,
        r=r,
        T=T,
```

```
        n_steps=n_steps,
        n_paths=200,
        rng=np.random.default_rng(seed + 1),
        return_paths=True,
        return_variance_paths=False,
    )

    path_path = output_dir / f"mc_paths_SPY_{expiry}_K{int(round(
        strike))}.png"
    plot_mc_paths_by_payoff_class(
        s_paths=sim_paths.s_paths,
        T=T,
        K=float(strike),
        output_path=path_path,
        expiry=expiry,
        iv_source_label=iv_source_label,
    )

    # 3) dt-Konvergenz
    dt_path = output_dir / f"dt_convergence_SPY_{expiry}.png"
    plot_dt_convergence(
        df_expiry=df_expiry,
        params=params,
        steps_list=[25, 50, 100, 200, 400],
        n_paths=5000,
        seed=seed,
        output_path=dt_path,
        iv_source_label=iv_source_label,
    )

    # 4) MC-Konvergenz
    mc_path = output_dir / f"mc_convergence_SPY_{expiry}.png"
    plot_mc_convergence(
        df_expiry=df_expiry,
        params=params,
        n_paths_list=[5000, 10000, 20000, 40000],
        n_steps=n_steps,
        seed=seed,
        output_path=mc_path,
        iv_source_label=iv_source_label,
    )
```

```
116
117     # 5) Varianz-Stabilitaet
118     v_path = output_dir / f"v_stability_SPY_{expiry}.png"
119     plot_v_stability(
120         v_t=result["v_t"],
121         output_path=v_path,
122         expiry=expiry,
123         iv_source_label=iv_source_label,
124     )
125
126     # 6) Calibration Fit Scatter
127     scatter_path = output_dir / f"calibration_fit_scatter_SPY_{
        expiry}.png"
128     plot_calibration_fit_scatter(
129         market_iv=df_expiry["iv_market"].to_numpy(dtype=float),
130         model_iv=model_iv,
131         output_path=scatter_path,
132         expiry=expiry,
133         iv_source_label=iv_source_label,
134     )
135
136     # 7) IV-Residuals vs Moneyness
137     residual_path = output_dir / f"iv_residuals_SPY_{expiry}.png"
138     plot_iv_residuals_vs_moneyness(
139         df_expiry=df_expiry,
140         model_iv=model_iv,
141         output_path=residual_path,
142         expiry=expiry,
143         iv_source_label=iv_source_label,
144     )
145
146     # 8) Preisfit inkl. Bid-Ask-Band
147     price_fit_path = output_dir / f"price_fit_bidask_SPY_{expiry
        }.png"
148     plot_price_fit_with_bid_ask(
149         df_expiry=df_expiry,
150         model_price=result["model_price"],
151         output_path=price_fit_path,
152         expiry=expiry,
153         iv_source_label=iv_source_label,
154     )
```

```
155
156     # 9) Martingale-Check ueber Zeit
157     martingale_path = output_dir / f"martingale_check_SPY_{expiry
            }.png"
158     plot_martingale_check(
159         params=params,
160         S0=S0,
161         r=r,
162         T=T,
163         n_steps=n_steps,
164         n_paths=max(4000, min(12000, n_paths_final)),
165         seed=seed + 17,
166         output_path=martingale_path,
167         expiry=expiry,
168         iv_source_label=iv_source_label,
169     )
170
171     # 10) Residual-Heatmap ueber mehrere Expiries
172     heatmap_path = output_dir / "residual_heatmap_SPY.png"
173     plot_residual_heatmap_across_expiries(
174         cleaned=cleaned,
175         params=params,
176         output_path=heatmap_path,
177         iv_source_mode=iv_source_mode,
178         steps_per_year=steps_per_year,
179         n_paths=max(2500, min(6000, n_paths_final // 2)),
180         seed=seed + 41,
181         max_expiries=6,
182     )
183
184     # 11) Out-of-sample Smile-Overlay
185     oos_path = output_dir / f"smile_overlay_oos_SPY_{expiry}.png"
186     oos_expiry = plot_out_of_sample_smile_overlay(
187         cleaned=cleaned,
188         params=params,
189         in_sample_expiry=expiry,
190         output_path=oos_path,
191         iv_source_mode=iv_source_mode,
192         n_paths=max(4000, min(12000, n_paths_final)),
193         steps_per_year=steps_per_year,
194         seed=seed + 59,
```

```
195             calibration_expiries=calibration_expiries,
196         )
```

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